

Name: \_\_\_\_\_

Date: \_\_\_\_\_

# ALGEBRAIC EXPRESSIONS, (USING AND WRITING SIMPLE FORMULAE)

LO: I can write and use simple algebraic formulae to solve problems.

## What is a formula?

A **formula** is a mathematical rule written with **letters and numbers** that shows the relationship between quantities.

**Remember:** The subject of a formula is the single letter on its own, usually on the left side of the equals sign.

## Quick examples

|   |                  |                             |
|---|------------------|-----------------------------|
| ● | $P = 2l + 2w$    | perimeter of a rectangle    |
| ● | $A = l \times w$ | area of a rectangle         |
| ● | $C = 5n + 10$    | cost formula                |
| ● | $P = l + w$      | missing the 2s              |
| ● | $A = 2(l + w)$   | that is perimeter, not area |

## Key idea

Identify the variables, write the rule using letters, then substitute values to evaluate.

$$C = 5n + 10$$

5 per item plus 10 fixed cost

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 - using a formula

Use  $P = 2l + 2w$  to find the perimeter when  $l = 8$  and  $w = 5$ .

Substitute:  $P = 2 \times 8 + 2 \times 5$

Calculate:  $P = 16 + 10$

$P = 26$

$$P = 26 \text{ cm}$$

Replace each letter with its value, then follow BIDMAS.

### Example 2 - writing a formula

A taxi charges £3 fixed plus £2 per mile. Write a formula for cost  $C$  for  $m$  miles.

Fixed charge = 3

Cost per mile =  $2m$

Combine:  $C = 2m + 3$

$$C = 2m + 3$$

Identify the fixed part and the variable part.

### Example 3 - evaluating

If  $C = 2m + 3$ , find  $C$  when  $m = 7$ .

Substitute  $m = 7$ :  $C = 2 \times 7 + 3$

$C = 14 + 3$

$C = 17$

$$C = £17$$

Multiply before adding (BIDMAS).

### Example 4 - two-variable formula

The area of a triangle is  $A = \frac{1}{2} \times b \times h$ . Find  $A$  when  $b = 10$  and  $h = 6$ .

Substitute:  $A = \frac{1}{2} \times 10 \times 6$

$A = \frac{1}{2} \times 60$

$A = 30$

$$A = 30 \text{ cm}^2$$

Multiply base by height first, then halve.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Substituting into formulae

Use  $P = 2l + 2w$ . Find  $P$  for each pair of values.

- a)  $l = 6, w = 4$
- b)  $l = 10, w = 3$
- c)  $l = 7, w = 7$
- d)  $l = 12, w = 5$

### 2. Cost formulae

A plumber charges  $C = 30 + 20h$  pounds for  $h$  hours. Find  $C$ .

- a)  $h = 2$
- b)  $h = 5$
- c)  $h = 0.5$
- d)  $h = 8$

### 3. Writing formulae

Write a formula for each situation.

- a)  $n$  books at £4 each, total cost  $T$
- b)  $p$  pencils at 30p each plus 50p bag, total  $C$
- c) Perimeter of equilateral triangle side  $s$
- d) Weekly pay: £200 plus £15 per overtime hour  $h$

### 4. Word problems

Write and use a formula to answer each question.

- a) Gym costs £25/month + £5 joining fee. Cost for  $m$  months?
- b) Using your formula, find cost for 6 months.
- c) Rectangle:  $l = 9, w = 4$ . Find area using  $A = lw$ .
- d) Triangle:  $b = 12, h = 8$ . Find  $A = \frac{1}{2}bh$ .

### 5. Mixed formulae

Evaluate each formula

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) In  $C = 5n + 10$ ,  $C$  means the cost ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b)  $2l$  means  $2 + l$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c)  $P = 2l + 2w$  gives the perimeter of a rectangle ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d)  $A = lw$  gives the perimeter ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e) Substitution means replacing letters with numbers ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A mobile phone contract costs **£15 per month** plus **£0.05 per text**. Write a formula for the monthly cost  $C$  in terms of the number of texts  $t$ . Find the cost if you send **200 texts**.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) The formula for the area of a trapezium is  $A = \frac{1}{2}(a + b)h$ . A trapezium has parallel sides  $a = 8$  cm and  $b = 14$  cm, and height  $h = 6$  cm. Find its area. Then rearrange to find  $h$  if  $A = 55$ ,  $a = 6$ ,  $b = 16$ .

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

